

APPALA LOHITH VENKATA GUPTHA

+91 9121946531 | lohi.valli@gmail.com | LinkedIn | GitHub | LeetCode

PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience designing and developing scalable full-stack web and mobile applications using Java, Spring Boot, React.js, Node.js, and Flutter. Passionate about designing reliable software, solving complex engineering problems, and building high-quality applications using object-oriented programming, RESTful APIs, and modern software engineering practices. Proven ability to collaborate in Agile teams, quickly learn new technologies, and deliver performance-focused solutions.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript (ES6+), C++, Dart

Software Engineering Fundamentals: Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Development Lifecycle (SDLC), Unit Testing, Debugging, Code Reviews, Agile Methodology, Scalability, Performance Optimization, Reliability

Backend & APIs: Spring Boot, Spring MVC, Node.js, Express.js, RESTful API Design, JWT Authentication, Role-Based Access Control

Frontend & Mobile: React.js, HTML5, CSS3, Responsive UI Design, Flutter

Databases: MongoDB, PostgreSQL

Cloud: AWS S3, AWS CloudFront

Tools & Platforms: Git, GitLab, Version Control, Linux, Postman, VS Code

Core Concepts: Full-Stack Development, Microservices Basics, Service-Oriented Architecture, Requirements Gathering, Cross-Functional Collaboration

Soft Skills: Problem Solving · Analytical Thinking · Time Management · Clear Communication · Adaptability

EXPERIENCE

B2B Infotech Pvt. Ltd.

Trainee Software Employee

Vijayawada, India

Feb 2026 – Present

- Applied software engineering principles to maintain and enhance full-stack web applications using the MERN Stack (MongoDB, Express.js, React.js, Node.js), ensuring high availability and performance in production.
- Designed and implemented RESTful APIs to enable efficient communication between frontend and backend services, improving data throughput while reducing latency.
- Developed cross-platform Flutter applications for Android and iOS by rapidly adopting Flutter, ensuring consistent UI/UX and reliable performance across devices.
- Designed and managed MongoDB database schemas for products, users, orders, and inventory, supporting scalable data operations.
- Implemented secure user authentication, authorization, and session management, strengthening application security and access control.
- Performed testing, debugging, deployment, and defect resolution to improve application reliability, maintainability, and production readiness.
- Optimized API response times and database queries to improve scalability and performance for concurrent users.
- Collaborated with product stakeholders and cross-functional teams following Agile practices, including sprint planning, code reviews, and daily standups.
- Used Git and GitLab for version control, collaborative development, and code review workflows.

Blackbucks Pvt. Ltd.

AI/ML Intern

Hyderabad, India

May 2025 – July 2025

- Engineered a fuel efficiency prediction system using Linear Regression, Random Forest, XGBoost, and Keras-based Deep Neural Networks, improving mileage estimation accuracy through systematic model comparison and evaluation.
- Designed and implemented end-to-end data pipelines covering preprocessing, feature engineering, and statistical evaluation on real-world datasets.
- Implemented automated model versioning and report generation, standardizing reproducibility and reducing manual tracking effort across 4+ model variants.
- Applied supervised learning and regression modeling techniques to deliver a production-ready predictive analytics solution aligned with business requirements.

Aimers Society

Software Development Intern

Vijayawada, India

May 2024 – July 2024

- Designed and developed an AI-powered conversational chatbot using Generative AI and NLP, achieving 85–90% response accuracy through iterative testing across diverse user query scenarios.
- Integrated the chatbot with backend services via RESTful APIs, improving data exchange efficiency and reducing average response latency.
- Built Power BI dashboards to monitor 5+ KPIs, including query trends, system efficiency, and user engagement, supporting data-driven product decisions.
- Implemented intent detection and response optimization mechanisms, improving context-awareness and reducing fallback responses in production testing.

PROJECTS

Anti-Proxy Smart Appearance Monitoring System

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Python, FastAPI, React.js, OpenCV, YOLOv8, InsightFace, PostgreSQL (Supabase), JWT, REST APIs, Git

- Engineered an AI-powered attendance system using YOLOv8 for real-time face detection and InsightFace for face verification, preventing proxy attendance and ensuring identity authenticity.
- Designed and developed a scalable FastAPI backend that enabled real-time communication between the frontend, AI inference pipeline, and PostgreSQL database.
- Implemented image preprocessing and face alignment using OpenCV to improve recognition accuracy and reliability across varied lighting and pose conditions.
- Designed a secure, multi-role access control system with JWT-based authentication, enforcing distinct privilege levels for administrators, teachers, and students.
- Architected the system for concurrency and real-time reporting, ensuring reliable performance under simultaneous user load.

Hospital Management System

| GitHub

Java, Spring Boot, REST APIs, PostgreSQL, AI Integration

- Architected a scalable hospital management platform unifying doctor, nurse, and patient workflows into a single full-stack system using Spring Boot and RESTful API design.
- Developed role-based access control (RBAC) and a secure reporting module enabling real-time access to downloadable medical records, reducing manual documentation overhead.
- Integrated an AI-powered virtual assistant for patient engagement, improving system navigation and reducing support queries through automated contextual responses.
- Applied service-oriented architecture principles to ensure modularity, scalability, and maintainability across appointment, prescription, and EMR management modules.

EDUCATION

PSCMR CET

Vijayawada, India

Bachelor of Technology – Computer Science (Artificial Intelligence)

2022 – 2026

- GPA: 7.82/10

Sri Bhavishya Educational Academy

Vijayawada, India

Intermediate – PCM

2020 – 2022

- Percentage: 68.5%

Nellore Ravindra Bharati School

Vijayawada, India

SSC – Matriculation

2019 – 2020

- GPA: 9.85/10

CERTIFICATIONS

Full Stack Web Development – MEAN, MERN, MongoDB (Udemy) · Backend Development with Node.js (HackerRank)
· Front-end Development – HTML, JavaScript · Artificial Intelligence · Python (Basic) · Database Management Systems